A COMPLETE SNOWFLAKE COUNTEREXAMPLE TO THE BOREL-REPRESENTATIVE PROBLEM FOR NEWTONIAN FUNCTIONS

LITERATURE-IMPLIED ANSWER PACKET FOR ARXIV:1509.02326

Abstract. Björn–Björn–Malý asked whether every bounded Newtonian function on a metric space with a positive complete Borel measure finite on balls has a Borel representative. We give a negative answer for every $1 \leq p < \infty$. Snowflake the interval by $d_\alpha(x, y) = |x - y|^\alpha$, $0 < \alpha < 1$, and equip it with an explicit complete finite non-Borel-regular extension of Lebesgue measure. There are no nonconstant rectifiable curves, so $N^{1,p} = L^p$ and $C_p(E) = \mu(E)$ for measurable E. The characteristic function of a Bernstein set then has no Borel representative. More generally, on this snowflaked interval all Newtonian functions have Borel representatives if and only if the measure is Borel regular, by a 2024 theorem of Alvarado–Gorka–Slabuszewski. We classify the result as literature-implied because that theorem supplies the general regularity criterion, although it does not mention the Newtonian problem.

Status. Literature-implied full negative answer, with an explicit construction and complete proof supplied for human review.

1. The source problem and the later criterion

The source is A. Björn, J. Björn, and J. Malý, *Quasiopen and p-path open sets, and characterizations of quasicontinuity*, arXiv:1509.02326 [1]. Under the standing assumptions $1 \leq p < \infty$ and a metric space carrying a positive complete Borel measure finite on balls, their Open Problem 5.3 asks:

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> The following open problems are natural in view of Corollary 5.2 and the comment after it.
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> **Open problem 5.3.** Does every (bounded) $u \in N^{1,p}(X)$ have a Borel representative?

Source evidence: Open Problem 5.3, PDF page 13 of arXiv:1509.02326.

Here a representative of $u \in N^{1,p}(X)$ means a function equal to u quasieverywhere, with respect to the source's Sobolev capacity.

A later theorem supplies the measure-theoretic dividing line. In arXiv:2410.21434, Alvarado–Gorka–Slabuszewski prove that, for a metric space with a nonnegative measure defined on a sigma-algebra containing the Borel sets and finite on balls, the following are equivalent: the measure is Borel regular, and every measurable function has a Borel representative almost everywhere [2].

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turns out that such assumptions are unnecessarily strong, in general. Indeed, for σ-finite, Borel regular measures on a topological space, every measurable function has a Borel representative; see, for example, [9, Proposition 3.3.23].[1] We show in Example 5.2 that the σ-finite assumption cannot be omitted, in general, for this result to hold true. Bringing this topic to a natural conclusion, in Theorem 5.1, we prove that Borel regularity is also necessary for the existence of Borel representatives, even for non-σ-finite measures.

As a corollary of our characterizations in Theorems 3.6, 4.4, and 5.1, we prove that under certain mild conditions on the topological measure space, Borel regularity fully characterizes both the weak and strong versions of Lusin's theorem and the existence of Borel representatives; see Theorem 6.1. It is important to note that the conditions on the topological measure space in Theorem 6.1 occur naturally. In fact, by specializing Theorem 6.1 to the setting of metric spaces equipped with a Borel measure, we obtain the following characterization. Here, and thereafter, we let $\overline{\mathbb{R}} := [-\infty, \infty]$.

Theorem 1.1. *Let (X, d) be a metric space and suppose that μ is a nonnegative measure that is defined on a σ-algebra which contains all Borel sets in X and has the property that every ball has finite measure. Then, the following statements are equivalent.*

(a) The measure μ is Borel regular, i.e., for every measurable set $E \subseteq X$, there exists a Borel set $B \subseteq X$ such that $E \subseteq B$ and $\mu(B) = \mu(E)$.

(b) Each measurable function has a Borel representative, i.e., for every measurable function $u : X \to \overline{\mathbb{R}}$, there exists a Borel set $N \subseteq X$ and a Borel function $f : X \to \overline{\mathbb{R}}$ such that $\mu(N) = 0$ and $f = u$ pointwise in $X \setminus N$.

Supporting evidence: Theorem 1.1(a)–(b), PDF page 2 of arXiv:2410.21434.

That paper does not cite the source problem and does not discuss Newtonian spaces. The reduction below makes its relevance exact and also gives a fully explicit witness.

2. The snowflake reduction

Fix $0 < \alpha < 1$ and put

$$X = [0, 1], \qquad d_\alpha(x, y) = |x - y|^\alpha.$$

This metric induces the usual topology. Thus X is compact, complete, and separable, and its Borel sets are the usual Borel subsets of the interval.

Lemma 1. *There is no nonconstant rectifiable curve in (X, d_α).*

Proof. Let $\gamma : [a, b] \to X$ be continuous and nonconstant. Choose $s < t$ such that $\gamma(s) \neq \gamma(t)$, reverse the parameter if necessary, and write $x = \gamma(s) < y = \gamma(t)$ and $L = y - x > 0$. For an integer $n \geq 1$, continuity and the intermediate value theorem give successive first hitting times

$$s = t_0 < t_1 < \cdots < t_n \leq t, \qquad \gamma(t_j) = x + \frac{jL}{n}.$$

The d_α-length of γ is therefore at least

$$\sum_{j=1}^{n} d_\alpha(\gamma(t_{j-1}), \gamma(t_j)) = n \left(\frac{L}{n}\right)^\alpha = L^\alpha n^{1-\alpha}.$$

This tends to infinity, so γ is not rectifiable. □

Let μ be any complete finite Borel measure on this space, in the source's sense that its domain sigma-algebra contains all Borel sets. Because upper gradients in the source are tested on non-constant compact rectifiable curves, Lemma 1 makes the zero function an upper gradient of every measurable function.

Proposition 2. *For every $1 \le p < \infty$,*

$$N^{1,p}(X) = L^p(X,\mu), \qquad \|f\|_{N^{1,p}} = \|f\|_{L^p}.$$

Moreover, for every measurable $E \subseteq X$,

$$C_p(E) = \mu(E).$$

Consequently two measurable Newtonian functions agree quasieverywhere if and only if they agree almost everywhere.

Proof. The first assertion follows immediately from the zero upper gradient and the definition of the Newtonian norm. If $h \in N^{1,p}(X)$ satisfies $h \ge 1$ on E, then

$$\|h\|_{N^{1,p}}^p = \int_X |h|^p \, d\mu \ge \mu(E),$$

so $C_p(E) \ge \mu(E)$. Conversely $h = \mathbf{1}_E$ is measurable, lies in L^p because μ is finite, has upper gradient zero, and is admissible; hence $C_p(E) \le \mu(E)$. The last assertion follows by applying the identity to the measurable discrepancy set. □

In particular, the later Borel-regularity theorem immediately yields the following sharp statement for this reduction class.

Corollary 3. *On the snowflaked interval above, every function in $N^{1,p}(X)$ has a Borel representative for every $1 \le p < \infty$ if and only if μ is Borel regular.*

Proof. If μ is Borel regular, Theorem 1.1 of [2] gives a Borel almost-everywhere representative of every measurable L^p function, and Proposition 2 upgrades almost-everywhere equality to quasieverywhere equality. Conversely, if every Newtonian function has a Borel representative, then every bounded measurable function does, because $\mu(X) < \infty$ and $N^{1,p} = L^p$. In particular every characteristic function has a Borel representative; equivalently, one may apply the implication from Borel representatives to Borel regularity in [2]. □

3. An explicit complete nonregular measure

We now construct the promised measure rather than merely invoking the existence of a non-Borel-regular one. Recall that a *Bernstein set* $A \subseteq [0,1]$ is a set such that both A and A^c meet every nonempty perfect subset of $[0,1]$. Such sets exist by the classical transfinite construction: enumerate the nonempty perfect sets and choose two fresh points from each, putting one into A and reserving the other for A^c.

Neither A nor A^c contains a Lebesgue-measurable set of positive measure. Indeed, every positive-measure Lebesgue set contains a nonempty perfect set, which would contradict one of the two defining intersection properties. Thus

$$\lambda_*(A) = \lambda_*(A^c) = 0. \tag{1}$$

Let $\mathcal{L}$ be the Lebesgue sigma-algebra and define

$$\Sigma = \{(B \cap A) \cup (C \cap A^c) : B, C \in \mathcal{L}\}.$$

This is a sigma-algebra: complements and countable unions are computed coordinatewise on A and A^c. It contains $\mathcal{L}$, because $D = (D \cap A) \cup (D \cap A^c)$ for every $D \in \mathcal{L}$. Define

$$\mu\big((B \cap A) \cup (C \cap A^c)\big) = \frac{\lambda(B) + \lambda(C)}{2}. \tag{2}$$

Lemma 4. *Formula (2) is well-defined and gives a complete probability measure on Σ. Its restriction to $\mathcal{L}$ is Lebesgue measure.*

Proof. Suppose two pairs (B, C) and (B', C') represent the same set. Then $B \triangle B' \subseteq A^c$ and $C \triangle C' \subseteq A$. These symmetric differences are Lebesgue measurable, so (1) says that both are null. The right side of (2) is therefore independent of the representation.

For countable additivity, let represented sets E_n be pairwise disjoint, with coordinates (B_n, C_n). For $n \neq m$, the measurable intersection $B_n \cap B_m$ is contained in A^c, and $C_n \cap C_m$ is contained in A. Both intersections are null by (1). Thus each coordinate family is pairwise disjoint modulo null sets, and Lebesgue countable additivity applied to the coordinatewise union proves countable additivity of μ.

If $E = (B \cap A) \cup (C \cap A^c)$ is μ-null, then B and C are Lebesgue null. Every subset of E splits into a subset of $B \cap A$ and a subset of $C \cap A^c$; these two subsets are Lebesgue measurable by completeness of Lebesgue measure and hence give a member of Σ. Therefore μ is complete. Finally, a Lebesgue set D has the representation $(B, C) = (D, D)$, so $\mu(D) = \lambda(D)$. In particular $\mu(X) = 1$. $\square$

This μ is a positive complete Borel measure finite on balls: it contains the usual Borel sigma-algebra, agrees there with Lebesgue measure, and every nonempty open ball has positive finite measure. It is not Borel regular. Indeed, $\mu(A) = 1/2$, while every Borel $D \supseteq A$ has $D^c \subseteq A^c$. By (1), $\lambda(D^c) = 0$, whence $\mu(D) = \lambda(D) = 1$.

The same computation gives the stronger identity

$$\mu(A \triangle D) = \frac{\lambda(D^c) + \lambda(D)}{2} = \frac{1}{2} \qquad \text{for every Borel } D \subseteq X. \tag{3}$$

4. The full negative answer

Theorem 5. *For every $1 \leq p < \infty$ there is a compact complete metric space X with a finite positive complete Borel measure μ, and a bounded function $u \in N^{1,p}(X)$, such that u has no Borel representative. Hence the answer to Open Problem 5.3 of arXiv:1509.02326 is negative, even simultaneously for all finite p on one metric measure space.*

Proof. Use the snowflaked interval and the measure of Lemma 4, and put $u = \mathbf{1}_A$. This function is bounded and measurable. Since $\mu(X) = 1$, Proposition 2 gives $u \in N^{1,p}(X)$ for every finite p.

Suppose a Borel function v represented u. The Borel set $D = \{x : v(x) > 1/2\}$ would satisfy

$$\{u \neq v\} \supseteq A \triangle D.$$

In particular (3) gives $\mu(\{u \neq v\}) \geq 1/2$. Proposition 2 then yields

$$C_p(\{u \neq v\}) = \mu(\{u \neq v\}) \geq \frac{1}{2} > 0,$$

contradicting equality quasieverywhere. Thus no Borel representative exists. $\square$

Remark 6 (Provenance)**.** Theorem 5 is filed as a literature-implied answer, not as a novel counterexample. The general equivalence between Borel regularity and Borel representatives is already Theorem 1.1 of [2]; the snowflake collapse, explicit Bernstein-set extension, and connection to the 2015 Newtonian open problem are supplied here. The supporting paper contains no citation to the source paper and no occurrence of "Newtonian" or "quasiopen," so it does not appear to advertise this consequence.

References

[1] A. Björn, J. Björn, and J. Malý, *Quasiopen and p-path open sets, and characterizations of quasicontinuity*, Nonlinear Analysis 138 (2016), 224–237; arXiv:1509.02326.

[2] R. Alvarado, P. Górka, and A. Słabuszewski, *Borel regularity is equivalent to Lusin's theorem and the existence of Borel representatives*, arXiv:2410.21434 (2024).